

Cognizant Digital Nurture 5.0

ReactJS Hands-On Exercise 7 - Props in React

Objective

The objective of this exercise is to understand and implement Props in React. The application demonstrates how data can be passed from a parent component to a child component using props. It also helps in understanding the difference between State and Props and the usage of ReactDOM.render().

Theory

What are Props?

Props (Properties) are read-only inputs used to pass data from one React component to another.

Features:

- Props are immutable.
- Used for communication between parent and child components.
- Make components reusable.
- Passed as attributes while rendering components.

Default Props provide default values when no value is supplied by the parent component.

Difference Between State and Props:

Props: Read-only, Parent supplied.

State: Mutable, Component managed.

ReactDOM.render() renders a React component into the HTML DOM.

Prerequisites

Node.js

npm

Visual Studio Code

ReactJS

Implementation Steps

1. Create a React application using `npx create-react-app shoppingapp`.
2. Navigate to the project folder.
3. Create `Cart.js`, `OnlineShopping.js`, `App.js` and `index.js`.
4. Create the Cart class.

5. Create an array of shopping items.
6. Pass shopping items using Props.
7. Display data using map().
8. Run using npm start.

Source Code

Cart.js:

```
export default class Cart {  
  
  constructor(itemName, price) {  
  
    this.itemName = itemName;  
  
    this.price = price;  
  
  }  
  
}
```

OnlineShopping.js:

```
import React, { Component } from "react";  
  
import Cart from "../Cart";  
  
class OnlineShopping extends Component {  
  
  carts = [  
  
    new Cart("Laptop", 65000),  
  
    new Cart("Mobile", 25000),  
  
    new Cart("Keyboard", 1200),  
  
    new Cart("Mouse", 800),  
  
    new Cart("Headphones", 2000)  
  
  ];  
  
  render() {  
  
    return (  
  
      <div  
  
        style={{  
  
          margin: "30px",
```

```
    fontFamily: "Arial"

  }}

  >

  <h1>Online Shopping</h1>

  <table

    border="1"

    cellPadding="10"

    cellSpacing="0"

  >

    <thead>

      <tr>

        <th>Item Name</th>

        <th>Price (₹)</th>

      </tr>

    </thead>

    <tbody>

      {this.carts.map((item, index) => (

        <tr key={index}>

          <td>{item.itemName}</td>

          <td>{item.price}</td>

        </tr>

      ))}

    </tbody>

  </table>

</div>
```

```
);  
}  
}  
  
export default OnlineShopping;
```

App.js :

```
import React from "react";  
  
import OnlineShopping from "./OnlineShopping";  
  
function App() {  
  
  return (  
  
    <div>  
  
      <OnlineShopping />  
  
    </div>  
  
  );  
}  
  
export default App;
```

index.js :

```
import React from "react";  
  
import ReactDOM from "react-dom/client";  
  
import App from "./App";  
  
const root = ReactDOM.createRoot(  
  
  document.getElementById("root")  
  
);  
  
root.render(  
  
  <React.StrictMode>  
  
    <App />  
  
  </React.StrictMode>  
  
);
```

Optional index.css :

```
body {  
    margin: 0;  
    padding: 20px;  
    font-family: Arial, Helvetica, sans-serif;  
    background-color: #f5f5f5;  
}  
  
h1 {  
    color: #0d6efd;  
}  
  
table {  
    border-collapse: collapse;  
    background: white;  
}  
  
th {  
    background-color: #0d6efd;  
    color: white;  
}  
  
th,  
td {  
    padding: 10px 20px;  
    text-align: center;  
}
```

Output Screenshot:



Item Name	Price (₹)
Laptop	65,000
Mobile	25,000
Keyboard	1,200
Mouse	800
Headphones	2,000

Learning Outcome

Learned Props, class components, `map()`, `ReactDOM.render()`, and dynamic data rendering.

Conclusion

This exercise demonstrates passing data using Props and rendering reusable React components.